3
Palladium Membranes

Stephen N. Paglieri

3.1
Introduction

Metal membranes that are composed principally of palladium alloys are used to purify hydrogen. Although tubular and foil membrane units are commercially available, several types of composite membranes are currently under development to minimize the use of palladium, a precious, platinum-group metal.

The demand for pure hydrogen is growing. Approximately 22% of all in-plant energy use or 4.5 quadrillion Btu per year (quads yr^{-1}) is consumed annually by separations [1]. Palladium membranes have the potential to reduce capital investment and energy consumption in industry by extracting hydrogen from hydrocarbon streams, for example. Palladium membranes may play an important role in the development of more environmentally sustainable energy systems, such as fuel cells for vehicle drives and distributed power [2, 3]. As part of process intensification, palladium membrane reactors merge reaction and separation to increase efficiency.

To compete successfully with conventional hydrogen separation technologies, such as pressure swing adsorption or polymer membranes, or to be used in a membrane reactor successfully, palladium membranes must possess the following characteristics [4]:

- Constant high hydrogen flux and permselectivity
- Chemical resistance to common gas stream components such as steam, and carbon and sulfur compounds
- Long lifetime (>10 000 h)
- Durable (to withstand thermal cycling)
- Cost effective and straightforward to fabricate
- Ability to be assembled into compact (high surface-to-volume ratio) modules with membranes that are easy to seal and replace or repair

Recent progress in addressing these problems in palladium membrane technology involves:

Nonporous Inorganic Membranes.
Edited by Anthony F. Sammells and Michael V. Mundschau
Copyright © 2006 WILEY-VCH Verlag GmbH & Co. KGaA, Weinheim
ISBN: 3-527-31342-7

1. The fabrication of composites to decrease cost and increase hydrogen flux by curtailing the amount of palladium required.
2. The realization of more chemically and thermally stable membranes through the investigation of palladium alloys and modified porous supports.

For example, effective intermediate layers have been developed that create a smoother support surface to minimize defect density and reduce metallic interdiffusion. The following sections of this chapter examine the history, background, and applications of palladium membranes, including recent advances in fabrication methods and performance.

3.2 History and Applications

Hydrogen was first permeated through palladium in 1863 by Sainte-Claire Deville and Troost; the ability of palladium to absorb hydrogen to several hundred times its volume at room temperature was recorded by Sir Thomas Graham in 1866 [5]. The first patent on hydrogen purification using palladium was issued to Snelling in 1916 [6]. Until the 1950s, palladium membranes appeared to be used primarily as a leak for pure hydrogen [7]; they were used later in chromatography [8–10] and other scientific endeavors [11–13]. Palladium membrane modules experienced more widespread use in the 1960s with the appearance of small-scale on-site hydrogen generation units for laboratory, military, and industrial uses [14–22]. Some of these units reformed a liquid fuel to obtain hydrogen [23–27]. Other membranes were developed to conduct chemical reactions [28].

Industrial applications for palladium membranes include compound semiconductor manufacturing where ultra-high purity hydrogen is required to minimize the formation of defects during metal-organic chemical vapor deposition (MOCVD) [29]. Palladium alloy membranes are used to take already pure hydrogen to parts-per-billion (ppb) levels of impurities, and commercial systems are currently available for hydrogen purification or generation [30–32]. Palladium membranes are also used to recover hydrogen isotopes through separation from gas mixtures [33–36]. In theory, multi-staged palladium membranes can be used to separate hydrogen isotopes from each other because the isotope separation factor of palladium is relatively high [13, 37–41]. A hydrogen-permeable membrane can also improve the efficiency of a high-temperature chemical heat pump [42].

There are many opportunities to recover hydrogen. Sour crude oil is subjected to hydrotreating, cracking and desulfurization [43]. Hydrogen that can be separated from catalytic cracking and hydroforming process streams is currently burned or flared [1]. Other high-pressure streams containing hydrogen include purge gas streams from ammonia or methanol synthesis. Several thorough reviews are available for more in-depth coverage of palladium membrane technology fabrication and uses [35, 41, 44–48].

A palladium membrane reactor can be used to couple hydrogenation and dehydrogenation reactions, or drive a reaction towards completion (Le Chatelier–Braun's principle) by adding or extracting hydrogen in a liquid or gaseous phase reactor [49–51]. Membranes are also used to study chemical reaction mechanisms [52–54]. The hydrogen that wells-up to the surface of the membrane has a higher degree of activity, so the membrane can also serve as a catalyst. Selective hydrogenation on certain palladium alloys or membranes with special surface coatings minimizes waste of valuable feedstock in the fine chemical or pharmaceutical industries [55]. Membrane reactors can operate at reduced temperatures while maintaining the same conversion rate and yield as a conventional reactor, thus lowering the quantity of catalyst required and allowing reactors to be constructed with less expensive materials [56–59]. A membrane reactor can also concentrate carbon dioxide in the high-pressure retentate, expediting carbon sequestration [60]. Hydrogen isotopes have been extracted from radioactive waste using a palladium membrane reactor [61–63]. More information on the wider field of palladium membrane reactors can be found in other sources [4, 28, 64–67].

3.3 Effect of Impurities

Oxygen, nitrogen, carbon, and hydrogen all affect the structure of palladium. Oxygen induces multiple effects. Exposing palladium to oxygen creates palladium oxide and promotes the formation of surface roughness or hillocks that can increase hydrogen permeability but can also cause the opening of pores or microvoids in thin films as revealed by atomic force microscopy (AFM) studies [68–72]. Oxygen also removes surface impurities (or contaminants in the pores of the support) such as carbon that interfere with hydrogen adsorption [72, 73]. Because the effects of oxygen are mostly beneficial, air treatment is commonly used to activate membranes before permeability testing [74]. However, thorough purging with an inert gas is required to avoid an explosive mixture. The optimal air treatment conditions for a Pd–Ag/alumina composite membrane were determined to be 400 °C for 1.5 h [72]. In contrast to air, hydrogen exposure results in a gradual flux decrease because of its surface smoothing effect, likely through the facilitation of atomic mobility [70]. The kinetics for hydrogen adsorption can also be improved by etching [75] or treating the membrane with palladium or platinum black to increase the effective surface area.

Carbon can block permeation and create porosity at higher temperatures, which is detrimental to both membrane stability and permselectivity [76–79], especially in combination with oxygen [71]. Exposure to unsaturated hydrocarbons at elevated temperatures is particularly detrimental [77, 80]. The formation of carbon on both the membrane and catalyst is promoted in palladium membrane reactors because of the selective removal of hydrogen [81], which necessitates the study of membrane, reactant/product gas mixture, and specialized catalyst in concert [51, 82]. For example, a Pd75–Cu25 (all compositions in this chapter are given in

weight percent) membrane was found to possess much higher catalytic activity for the reverse water-gas shift reaction than pure palladium [83].

The importance of testing with gas mixtures was demonstrated when a feed of 50/50 H_2/N_2 resulted in a much lower permselectivity because of increased nitrogen permeation [84]. Formation of ammonia on the catalytic surface of the palladium possibly blocked hydrogen diffusion. Swelling in the presence of hydrogen may either reduce or enlarge the size of microscopic fissures along grain boundaries as permselectivity increases and decreases have both been observed [78, 84, 85]. Further investigation into this phenomenon is warranted because the results so far are somewhat inconclusive. Hydrogen does stimulate the co-permeation of carbon through thicker palladium alloy membranes when CO and CO_2 are available to dissociate on the membrane surface. Methane materializes downstream from the reaction of hydrogen with diffused carbon [86]. Installation of a methanator can form methane upstream that is not adsorbed on the membrane surface [87]. Oxygen and nitrogen have also been observed to co-permeate with hydrogen.

Surface restructuring in alloys takes place during heating and may result in surface segregation of an individual element, depending on the gas to which the membrane is exposed [80, 88, 89]. For example, after exposure to air at temperatures up to 650 °C, the surface of a Pd–Ag membrane was enriched in silver [72]. Annealing in CO or CO_2 at temperatures as low as 250 °C altered the surface topology of a Pd Cu foil by forming hillocks and defects in addition to causing segregation of palladium to the surface exposed to these gases [83]. Cold rolling causes deformation in the process of forming thin foils or tubes, creating stress and microstructural defects such as dislocations [90]. Annealing, even at temperatures as low as 250 °C, reduces strain and eliminates dislocations and grain boundaries that trap hydrogen and decrease the diffusivity [68, 90, 91]. Annealing may cause an increase in hydrogen permeability through a membrane with time, whereas surface segregation of elements from the membrane bulk, such as sulfur or carbon, may cause a loss of permeability [92].

Hydrogen sulfide (H_2S) and carbon monoxide (CO) gases are commonly encountered in membrane reactors and severely poison membranes. The Pd77–Ag23 alloy has limited tolerance to sulfur compounds [10] and pure palladium is destroyed by corrosive H_2S [93–95]. However, Pd–Cu and Pd–Fe alloys appear to be resistant to sulfur attack and even maintain some permeability during H_2S contact [70, 96–98]. Furthermore, the original permeability can be restored by treatment in air or steam [70, 76] or hydrogen [99]. Carbon monoxide and steam (according to some authors) also inhibit hydrogen permeability by being reversibly adsorbed on the surface, obstructing sites where hydrogen is adsorbed and dissociates [82, 92, 100–104]. Partially oxidized Pd–Al membranes display increased resistance to CO [103]. Permeation hindrance by CO, H_2S and steam is more pronounced at lower temperatures and diminishes at temperatures >400 °C [105]. Poisoning by carbon tetrachloride was reversed by steam [106].

Certain metals, such as mercury and germanium (a catalyst component), have been found to irreversibly poison the surface of palladium and should be avoided [74, 99, 106]. Palladium membranes may not be able to withstand direct contact

with certain catalysts that contain volatile or low-melting metals because of metallic diffusion onto the membrane surface. One solution is to place the membrane after the reactor, but this limits the benefit of *in situ* hydrogen extraction from the reaction zone. Even minimal surface contamination on either the upstream or downstream side of the membrane can inhibit flux by affecting the processes of adsorption or desorption.

3.4 Palladium Alloy Membranes

Palladium membranes must possess sufficient strength to operate under large pressure differentials at elevated temperatures for extended periods of time. Additional desirable characteristics include resistance to common contaminants such as sulfur and the ability to thermally cycle under hydrogen. The use of palladium alloys can help satisfy physical and chemical stability requirements.

Pure palladium embrittles too easily upon hydrogen exposure and must be alloyed to improve its material properties and suppress the formation of hydrides [107]. The lattice of palladium retains its face centered cubic (fcc) structure while it expands during hydride formation ($\alpha \rightarrow \beta$ phase transition), creating substantial stress and reducing ductility [108–110]. The hydrogen solubility in palladium decreases with increasing temperature, and the critical temperature for formation of β-phase palladium is ~300 °C. Therefore, exposing palladium to hydrogen below this temperature should be avoided to prevent warping and cracking. The addition of other elements to palladium can decrease the critical temperature for β-phase formation so that membranes can be operated at lower temperatures without failure by embrittlement.

The industry standard is the Pd77–Ag23 alloy contained in most commercially operating units [12]. The permeability of Pd77–Ag23 is up to 1.7 times the permeability of pure palladium at typical operating temperatures of 300–450 °C [107]. Although the Pd77–Ag23 alloy is less affected by carbon monoxide exposure [111], it does suffer from grain coarsening at higher temperatures [112]. Researchers have extended the service lifetime by creating ternary palladium alloys with elements such as ruthenium, yttrium, aluminum, iron, and rhodium to impart higher strength and reduce stress resulting from hydrogen dilatation while maintaining hydrogen permeabilities comparable to palladium [28, 35, 75, 97, 113–118]. Elements with higher melting points such as platinum, rhodium, and nickel increase resilience to alleviate fatigue during operation under pressure at high process temperatures [80, 119–121]. Capillary tubes (30–50 μm wall thickness) of palladium alloys can withstand large pressure differentials [113].

Alloys help alleviate the problems associated with flux reduction from impurity adsorption on the surface of palladium membranes [74, 122]. Steam, CO, and H_2S inhibit the hydrogen flux through palladium membranes by chemisorption on the surface. There is some indication that palladium alloys are affected less by CO poisoning [111, 117, 123–125]. Recently, work on Pd–Cu alloys has increased

because of their sulfur tolerance and potentially more favorable materials properties, such as resistance to hydrogen embrittlement [33, 70, 96, 126–131]. According to the phase diagram, around 42 wt.% copper Pd–Cu forms a body centered cubic (bcc) phase below 600 °C that is more permeable to hydrogen [132, 133] but less resistant to hydrogen sulfide than the fcc phase that forms at higher temperatures [96, 134]. The hydrogen permeability of the Pd–Cu alloy supposedly reaches a maximum at the composition Pd58–Cu42 and is similar to pure palladium [96, 107, 126, 132, 135]. Platinum is sulfur tolerant but its hydrogen permeability is an order of magnitude less than that of palladium [94].

3.5 Palladium Deposition Methods

Common palladium deposition methods include physical vapor deposition (PVD, including electron beam or laser evaporation and dc magnetron sputtering), chemical vapor deposition (CVD or MOCVD), electroplating, and electroless plating (EP). Each method has strengths and weaknesses; therefore, it is best to tailor the deposition method to the particular membrane support to develop an effective composite membrane. A clear understanding of the parameters that determine microstructual evolution during deposition is critical. For example, CVD and electroless plating can more easily penetrate porous materials, whereas PVD offers better alloy composition control. Novel deposition methods have been tried such as solvated metal atom [136] or electrochemical [137] deposition from a supercritical fluid, spray pyrolysis [138], high-velocity oxy-fuel spraying (HVOF) [84], and impregnation [139]. Because they have not been optimized, these latter methods generally require the deposition of relatively thick palladium layers to eliminate pinholes.

Thorough cleaning and the avoidance of any extraneous debris are essential in any membrane fabrication scheme. After EP, a composite membrane can be soaked in hot water or dilute ammonia to help remove any impurities trapped in the porous support that may be detrimental to the palladium film during high temperature operation [72, 73]. However, traces of impurities from the EP bath such as chlorine, sodium, and carbon, inevitably become incorporated into the metal film. Membrane defects can be a consequence of preparation conditions. Fabrication in a clean-room environment has resulted in increased permselectivity [140]. Porous stainless steel (PSS) must be cleaned and pickled before electrodeposition or else activated for electroless plating.

Although it is somewhat more complicated in terms of equipment, CVD can promote transport of metal directly into pores for deposition [141]. Precursors such as palladium(II) chloride, palladium(II) acetylacetonate [$Pd(acac)_2$], and palladium bis(hexafluoroacetylacetonato) [$Pd(hfacac)_2$] are volatilized and entrained to the heated reaction zone, where the material is preferentially deposited within the pores. Counterdiffusion of the reducing gas (hydrogen) can aid in depositing metal more precisely [142]. Alloys may be deposited from a single mixed source.

It is important to ensure the metal-organic precursor is completely reacted in order to avoid carbon incorporation into the film [143].

EP is an autocatalytic electrochemical reaction based on the mixed potential theory where metal ions and a reducing agent essentially react simultaneously on a surface to deposit the metal [144, 145]. EP offers the flexibility to apply very thin, conformal, and continuous coatings to a variety of porous or nonporous and metallic or insulating surfaces. Because EP is a catalytic process, it requires proper surface activation to provide full film coverage and adhesion, especially on porous supports. A palladium and tin method is very effective on inactive metals, ceramics and polymers, despite the possibility that residual tin under the palladium film may promote mechanical instability during operation at temperatures ≥500 °C [73, 146]. Consequently, a preferred activation method is dip-coating in palladium acetate or $Pd(acac)_2$ followed by reduction in hydrogen before plating [146, 147]. Alternatively, palladium deposited by CVD has been used to seed the porous surface to be plated [148, 149]. Other common aqueous reductants for both the formation of activation nuclei and use in electroless plating solutions are formaldehyde, hypophosphite, hydrazine, dimethylamine borane, and borohydride. Hydrazine is preferred for palladium because hypophosphite, dimethylamine borane, and borohydride co-deposit phosphoros and boron, respectively. Formaldehyde is an effective reducing agent for copper deposition [150].

The optimization of EP parameters based on a fundamental understanding of the process has been the focus of much study. For example, conducting the processes of both activating and plating while evacuating the opposite side of the membrane allows enhanced penetration into the pores [73, 81, 148, 149, 151]. Ultrasound also has a positive impact by thinning the boundary layer adjacent to the surface to be plated [148]. Agitation is important for dislodging bubbles that cling to the surface during deposition and may result in voids. Deposition rate is strongly influenced by temperature and concentration of metal, complexing agent, and reducer [144]. Bath constituents often include stabilizers and surfactants. Carrying out EP in stages is advantageous. Although dilute EP solution better penetrates pores to fill and then bridge them, it also forms more loosely packed palladium crystallites [148, 151]. To obtain the dense structure required for a continuous metal film that is impervious to gases other than hydrogen, concentrated EP solution is used after a cycle with dilute EP solution [73, 151]. However, if plating is conducted too quickly, the film generated is not dense and has poor adhesion. Furthermore, the EP solution may decompose spontaneously.

Composite membranes are now routinely fabricated using palladium alloys. Alloys can be created by sequential deposition of the constituents followed by annealing [150, 152–156] or co-deposited using electroplating or electroless plating [137, 147, 157–159], spin-coated inks [160], dual electron-beam PVD [161], or sputtering from a target of the desired composition. Sputter target equilibration before exposure to the surface to be coated is important for depositing an alloy composition that is the same as the target [162]. Temperature and deposition rate are important parameters that determine the structure of films created by PVD [163]. For uniform coverage during PVD, tubular supports must be rotated to prevent shadowing.

Forming a defect-free, homogenous alloy layer from sequentially deposited metals depends on the annealing conditions [68, 152, 156, 159, 164]. Alloying is faster at higher temperatures and for thinner films. However, depending on the metal film thickness and support composition, high temperatures at which pinholes form should be avoided. Co-deposition requires less soak time and milder temperatures to form a homogeneous alloy. For example, after co-deposition of a Pd77–Ag23 film using EP, homogeneity was achieved after only 4 h at 500 °C [147]. Metal atom mobility becomes more pronounced at temperatures greater than half the melting point, referred to as the Tammann temperature [164]. Annealing should be conducted around this temperature in the presence of hydrogen, which facilitates atomic mobility [165].

3.6 Membrane Characterization and Analysis

Volumetric steady-state gas permeation tests at elevated temperatures are typically used to characterize the performance of palladium membranes. Electrochemical methods are also effective, even at high temperatures [90, 166]. Permeation through palladium depends on a solution-diffusion mechanism, including the steps of chemisorption and dissociation into atoms, absorption into the metal, diffusion through the metal lattice, transfer from the bulk metal to the opposite side, and recombination into molecules for desorption [167, 168]. Diffusion of molecular hydrogen through boundary layers adjacent to the surface is also necessary.

The equation used to describe flux is derived from a combination of Fick's and Sieverts' laws where the difference between the square root of hydrogen partial pressure on the feed and permeate sides of the membrane creates the driving force for hydrogen flux [168, 169]:

$$J = \frac{P(p_{H_2,1}^n - p_{H_2,2}^n)}{l} \tag{3.1}$$

The flux, J (mol m^{-2} s^{-1}), is a function of the permeability, P (mol m^{-1} s^{-1} Pa^{-n}), l (m) is the membrane thickness, $p_{H_2,1}$ (Pa) is the partial pressure of hydrogen on the upstream or retentate side of the membrane, $p_{H_2,2}$(Pa) is the partial pressure of hydrogen on the downstream or permeate side of the membrane, and n is the power dependence of hydrogen permeation through the membrane. Pressure-normalized flux or permeance (mol m^{-2} s^{-1} Pa$^{-0.5}$) is defined as the ratio of P/l and is customarily used for composite configurations. Permeability is a function of hydrogen diffusivity and solubility [168]. For palladium, solubility decreases and diffusivity increases with temperature for an overall increase in permeation rate at higher temperatures. Generating Arrhenius plots of permeance versus inverse temperature can show total activation energies ranging from 1 to 33 kJ mol^{-1} (depending on film thickness, composition, and deposition method) with values around 15 kJ mol^{-1} most common [78, 165, 170, 171]. Activation energy also decreases as temperature increases.

Estimating n is dependent on the permeability measurement and is obtained using least-squares regression of the slope of a plot of hydrogen flux versus the square root of transmembrane pressure differential [169, 171]. Deviation of the intercept from the origin can indicate the degree of data divergence from the ideal [172]. According to Sieverts' law for a dilute solution, the exponent n is equal to 0.5 when permeation is controlled by atomic hydrogen permeation through the bulk metal [172, 173]. At higher temperatures where the solubility of hydrogen in palladium is lower, this condition is often satisfied. Values of $n > 0.5$ indicate either rate limitation by adsorption/desorption or a contribution from other transport mechanisms to hydrogen flux through the membrane. In general, n tends to increase when surface effects, such as hydrogen dissociation, become more dominant as films become thinner [149, 174]. A model based on rate equations for each step of the permeation process predicted that bulk diffusion should be the rate-determining step for films >1 μm thick above 300 °C [167], although $n = 0.5$ has been observed for thinner films [165]. At relatively low temperatures, surface processes become rate controlling and may limit permeance [175].

An essential membrane property is permselectivity for hydrogen. Higher levels of impurity in the product may be acceptable in certain processes, although a permselectivity of at least 50 is desired in a membrane reactor to obtain a worthwhile increase in conversion [77]. A room temperature inert-gas pressure-drop test can indicate if a composite membrane should be tested further. Ultra-thin palladium films often contain imperfections that decrease the hydrogen permselectivity. Hence, other permeation mechanisms can operate, such as viscous (Poisselle) flow, ordinary (Fickian) diffusion, and Knudsen diffusion [64, 73]. Knudsen diffusion is the prevailing leak mode because of the small defect size usually encountered in thin palladium films. Knudsen diffusion results in minimal permselectivity towards lighter gases, based on the ratio of square root of molecular weight. Surface diffusion through grain boundaries is proportional to p instead of $p^{0.5}$ [176]. The closer n is to 1, the greater the contribution from other diffusion mechanisms besides solution-diffusion. If viscous flow predominates, then the value of n can exceed 1.

Composite membranes also frequently have $n > 0.5$ from resistance to hydrogen flux through the underlying support. Palladium films supported on a porous substrate will have a tortuous path for diffusion away from the exit side of the palladium film with a fraction of that surface blocked by the nonporous areas, reducing the surface area for desorption [45, 177]. The nature of defects in the palladium film can be quantified by measuring the flux through the porous support before palladium deposition and comparing this value with the overall flux through the composite membrane [178].

According to Eq. (3.1), flux should be inversely proportional to metal thickness, hence the incentive (besides cost) to fabricate composite membranes with very thin palladium layers. Flux is generally not inversely proportional to thickness for composite membranes with very thin palladium layers [179]. Moreover, because flux depends on the difference in hydrogen partial pressure, a sweep gas may be employed on the permeate side or it may be kept in vacuo. However, evacuation is

probably too expensive for most industrial processes, and it must be pointed out that the hydrogen must be separated from any sweep gas. Steam sweep has been suggested because it is easily separated and may assist in conducting heat away from an exothermic reaction in a membrane reactor [180]. The sweep gas may be inert or reactive, such as oxygen [10] or CO [181]. Reactive sweep gas can supply heat along a membrane reactor [57]. At high membrane fluxes, concentration polarization can seriously diminish permeation through the membrane as the hydrogen must diffuse through a layer of other gas to reach the surface [103, 117]. Therefore, it is important to optimize the recovery ratio by adjusting the flow rates of the feed (retentate) and sweep gas (if used). More detailed models of hydrogen permeation divide the membrane into elements for numerical analysis [13].

Palladium composite membrane fluxes ranging between 1 and 1×10^{-2} mol m^{-2} s^{-1} have been measured (see the comprehensive table in Ref. [179]). However, the following must be considered when comparing membrane performance. Permeance is very sensitive to factors such as n, sweep gas flow rate, and permselectivity. Lower permselectivity caused by pinholes or defects is often not noticeable at low transmembrane pressure differentials [179, 182]. Flux saturation can actually occur at high pressures for very thin membranes [168]. However, ultra-thin membrane films may be damaged by excessive pressure differentials. Analysis of impurity flux through gas chromatography or mass spectroscopy may not reveal the flux of impurities as accurately as direct volumetric flow rate measurement induced by a pressure differential. Finally, membrane permeance is only one of several important membrane characteristics, including hydrogen permselectivity, palladium film thickness, and robustness.

Many analysis tools are now routinely used to characterize the physical and chemical properties of membranes. Physical structure or morphology and film thickness are typically resolved using scanning electron microscopy (SEM), transmission electron microscopy (TEM), AFM, and profilometry. The pore-size distribution of membranes can be evaluated by porosimetry using low-temperature nitrogen adsorption–desorption or mercury [148, 183, 184]. Chemical composition and grain structure are evaluated with X-ray diffraction (XRD), Auger electron spectroscopy (AES), X-ray photoelectron spectroscopy (XPS or ESCA), energy dispersive X-ray analysis (EDX), and electron probe microanalysis (EPMA). Inductively coupled plasma (ICP) may be used to determine film composition as well as thickness [73]. Rutherford backscattering spectroscopy (RBS) reveals the two-dimensional distribution of elements [185].

Microstructural analysis using SEM is requisite for examining surface details and features such as film pore structure and thickness or the crystallites formed by the EP surface activation process [81, 186]. AFM can be used to probe micro- to nanoscale surface topography and determine critical parameters such as surface roughness, R_a (arithmetical mean deviation of the profile) [70, 154, 187] or the size and distribution of surface activation particles [188]. Nanoparticles formed during surface activation [147, 189] and the dislocation substructure of palladium [90] have been studied by TEM. Optical or laser profilometry is used to quantify surface roughness or to measure film thickness on a flat surface [190].

AES and XPS are primarily surface chemical analysis techniques and are useful for inspecting for contaminants. When they are coupled with ion bombardment, depth profiles can be obtained to analyze for compositional uniformity or the extent of metallic interdiffusion [72, 88, 161, 191, 192]. XRD is indispensable for determining alloy composition and estimating grain size, and EDX can be used for checking composition and elemental mapping [149]. EPMA is especially useful for measuring the composition across thin films. Palladium film thickness, compositional uniformity, and the extent of anchoring into a support has been studied by RBS [185, 193].

3.7 Palladium Composite Membranes

To reduce the amount of palladium used in membranes and to increase hydrogen flux, the thickness of the palladium film must be decreased. This is accomplished by depositing thin films of palladium onto porous, hydrogen-permeable supports that provide mechanical strength without limiting mass transport at high transmembrane pressure differentials. Some of the original work on composite membranes involved electroless plating of palladium onto porous Vycor glass [194]. Later, metal supports such as porous nickel or porous stainless steel (PSS) [191] and ceramics such as alumina (Al_2O_3) and zirconia (ZrO_2) were used as substrates. Tubular ceramic membranes have withstood pressures up to 30 bar [180]. Palladium–ruthenium has been deposited onto anodic alumina that had a uniform surface of well characterized pores [195]. Asymmetric membranes with a thin, selective layer supported by the bulk of the membrane consisting of larger pores, diminish the flow resistance through the membrane. Tubular, commercially available, porous ceramic membranes are usually asymmetric and consist of multilayers built up on a macroporous substrate [190], but porous stainless steel is symmetric (i.e. has the same particle size throughout).

Methods have been developed to optimize pore filling during palladium deposition. A variation of EP takes place by circulating a more concentrated solution (such as sucrose) on the opposite side of the membrane during electroless plating. An osmotic pressure gradient is created that promotes diffusion of plating solution into the pores, resulting in a more compact, nonporous deposit that better infiltrates the support [196, 197]. Alternatively, by evacuating the opposite side of a porous support during plating, an equivalent pore-plugging effect can be achieved [159, 198]. Cross-flow CVD has also been used to plug pinholes effectively [141, 199].

Composite membranes must be sealed into a fitting or module for testing. Tubes and thick foils can be brazed effectively and are indefinitely robust ≤ 450 °C. Although braze durability has evolved, care still must be taken to avoid excessive thermal cycling under hydrogen, which creates stress and can ultimately lead to membrane failure. Evacuating or purging with inert gas during start-up and shutdown alleviates this problem. Membranes fabricated from foils or tubes usually

have nearly infinite permselectivity when sealed correctly, with the purity of the permeated hydrogen restricted only by the diffusion of impurities such as carbon through the metal lattice and grain boundaries [87, 200]. For thin foils or supported thin films, sealing can be accomplished using metal or graphite gaskets and a bull-nose fitting.

For ceramic composite membranes, softer graphite ferrules, gaskets, or packing material are frequently used on the glazed portion of the support. Direct contact between the palladium and graphite should be avoided because carbon is somewhat soluble in palladium and has been linked to permeability decrease and pinhole formation [77, 78, 80]. Porous ceramics are often glazed with a high-melting glass composite to form a gas-tight surface to seal against. Ceramic–metal composites (cermets) are being developed for attaching porous ceramics to manifolds because sealing is also a problem in high-temperature fuel cells [201]. A particular advantage of PSS is that it can be welded into a module more easily. Because very thin palladium films are easily punctured by the conventional sealing methods described above, processes for welding or diffusion-bonding thin palladium alloy foils into tubular form [202–204] or onto frames [205] have also been developed.

The pores of ceramic, silica, or zeolite membranes can be impregnated with palladium compounds that are then reduced to the metal [206, 207]. The increased permselectivity for hydrogen may be a result of partial pore plugging that decreases the flux of impurities and enhances surface flow of hydrogen through the membrane. This configuration may be less susceptible to hydrogen embrittlement because the expansion and contraction of isolated particles would create less stress than a continuous film. Hydrogen permselectivities of up to several hundred have been observed with metal–ceramic selective surface-flow membranes, and metals other than palladium are effective as well [208–210]. Carbon molecular sieve membranes doped with palladium nanoparticles have exhibited H_2/N_2 permselectivities >1000 [211]. Ceramic particles may also be ion-exchanged or impregnated with metal before sol–gel dip-coating or slip-casting onto a porous support for surface modification. The layer, pre-seeded with palladium crystallites, is catalytic for further electroless deposition of palladium via EP [212].

Metals with high hydrogen permeabilities such as vanadium, niobium, and tantalum can also serve as foil supports for thin palladium films. The palladium protects the foil from oxidation and catalyzes hydrogen absorption and desorption. This type of membrane configuration is described in Chapter 4. Composite membranes consisting of palladium-coated polymers have been fabricated and achieved moderate permselectivity (H_2/CO_2 >100) [213]. Apparently the polymer can be damaged during the sputtering deposition process, and it may also be difficult to obtain a leak-free seal around a palladium/polymer composite. Further investigation of palladium polymer structures is probably warranted. EP may be a superior deposition method for polymers. A palladium-coated fuel cell membrane can prevent methanol crossover, but this is still under development [214, 215].

3.8 Recent Advances

Industrial membrane separators must have high surface area (SA) to volume (V) ratios. Conventional flat and tubular configurations have SA/V ratios of 30 and 250 $m^2\ m^{-3}$, respectively [4, 216]. Therefore, membranes are being developed that address the requirement of assembly into compact, high-throughput modules. Spin-extruded ceramic hollow fibers (diameter < 1 mm) consisting of α-alumina coated with γ-alumina were coated with 2 to 3 μm thick layers of palladium [184]. A composite capillary membrane exhibited high hydrogen flux and stability for 800 h at 430 °C. The γ-alumina intermediate layer, with its homogeneously incorporated palladium particles, is believed to result in a denser palladium film because it is an excellent catalyst for EP. With its smooth surface (consisting of a narrow pore-size distribution centered around 5.7 nm), the intermediate layer enabled the deposition of a relatively defect-free 4 μm thick Pd76–Cu24 palladium layer so that a H_2/N_2 separation factor of >5000 was attained at 510 °C and a pressure differential of 350 kPa [217]. The hydrogen flux under these conditions was 0.19 mol $m^{-2}\ s^{-1}$. New membrane supports, such as hollow fibers composed of reaction-bonded alumina or coated by centrifugal slip-casting, may also prove to be amenable to the support of thin, flawless palladium films [190].

Large tubes (14 mm in diameter and 0.8 m long) have been coated with thin (3 to 5 μm thick) Pd77–Ag23 films using sequential EP. Figure 3.1 displays a bun-

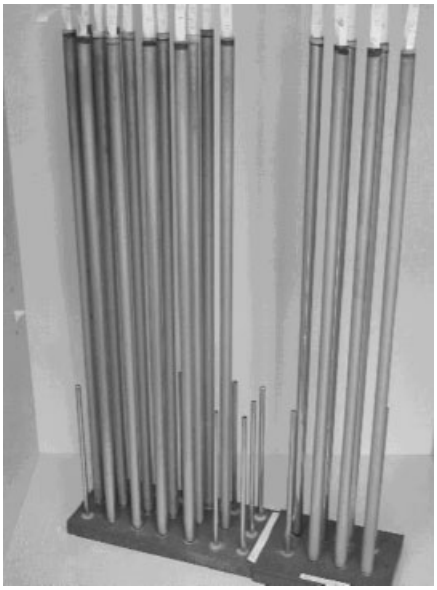


Figure 3.1 Bundle of 24 Pd–Ag coated membrane tubes with a length of 0.8 m for separator unit. (Photograph courtesy of ECN, The Netherlands)

dle of such tubes [182]. Modules with high surface area (~1 m^2), high permeability, and moderate permselectivity have been constructed. Permselectivity of >1000 was determined at lower pressure differentials, dropping to ~300 at higher pressures. Leak-free end seals that accommodate expansion and contraction with thermal cycling still remain a challenge. Preliminary results on multichannel monolithic membrane tubes with very high SA/V demonstrate that they can be coated with palladium [218].

Some efforts involve supporting ultra-thin (<1 μm) palladium films on various types of engineered porous supports. Perforated supports within modules have been fabricated and coated with thin layers of palladium alloys through micromachining and manufacturing techniques borrowed from the integrated circuit

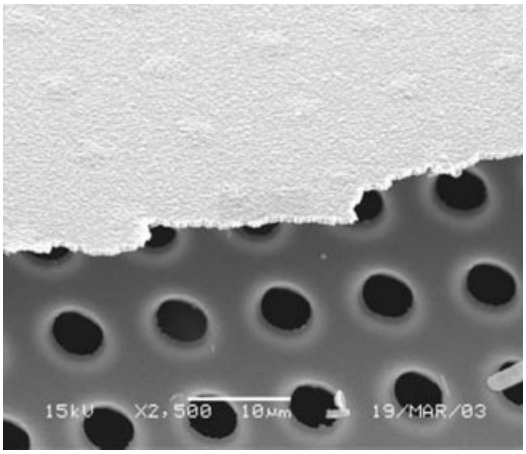


Figure 3.2 SEM micrograph of a Pd–Ag membrane layer deposited on top of a microsieve. The membrane layer was partly removed to show the microsieve.

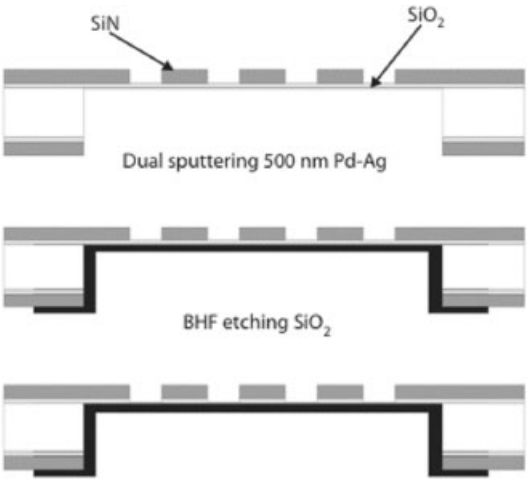


Figure 3.3 Fabrication process of the Pd–Ag membrane on a supporting microsieve.

industry (Fig. 3.2) [111, 161, 165, 219–222]. A typical microelectromechanical system (MEMS) membrane fabrication process to form a supported palladium membrane is shown in Fig. 3.3 [111]. For a 0.2 μm palladium film, fluxes up to 4 mol m^{-2} s^{-1} with a H_2/Ar permselectivity of ~1000 have been measured, although the mechanical strength of ultra-thin palladium films limits higher flux [111]. MEMS fabrication technology may be a feasible pathway towards the economic mass production of membrane systems with combined fuel processing capabilities.

Ultra-thin palladium alloy films have been deposited by PVD onto smooth substrates such as silicon wafers, removed, and wrapped around a PSS tube [223] or placed on a PSS disc [224]. The foil is sealed to the support by diffusion bonding at elevated temperature and pressure. Continuous foils as thin as 1 μm have been used to span the pores in the PSS support that are difficult to fill by conventional deposition methods. A mesh is used to protect the fragile surface from damage resulting from catalyst attrition. Additionally, membrane fabrication is amenable to scale-up through conventional PVD web-coating technology, in which large rolls of substrate are quickly coated under vacuum. The primary challenges of eliminating pinholes and forming hermetic seals remain.

Porous metal is a popular membrane support because of its sturdiness, close match to palladium in coefficient of thermal expansion, moderate cost, and ability to be welded into modules more easily. Commercially available PSS is often treated before palladium deposition to prevent metallic interdiffusion and create a smoother surface that enables coating with a thin, defect-free palladium film. The inherent surface roughness of PSS has been mitigated by shot peening [225], sintering on nickel microparticles, and coating with alumina or colloidal silica sol–gel [149, 226]. Composite palladium alloy membranes with higher hydrogen permselectivities and enhanced thermal stability were prepared on these improved surfaces. Of course, membrane stability then depends on the stability of the intermediate layer as well. Metallic interdiffusion that occurs between palladium and PSS at temperatures ≥450 °C [227] has been reduced by thin layers of oxide, nitride, or refractory metal on PSS [149, 156, 159, 164, 179, 191]. Palladium/PSS composite membranes with an oxide interlayer have demonstrated stability for more than 6000 h at 350–450 °C [156].

Complex substrate modifications involving intermediate layers and palladium alloy deposition methods are often required for superior membrane performance. Modification of a membrane support surface before palladium deposition by sintering on smaller particles can create a smoother surface with smaller pores, facilitating the deposition of a defect-free palladium layer. Nickel microparticles have been sintered together to form a porous support that was sputter-coated with palladium and then copper [118]. Thermal treatment at 700 °C for 1 h promoted reflow to create a durable, pinhole-free membrane with a Pd–Cu–Ni alloy film. In another case, starting with commercially available PSS with a 0.5 μm particle filtration cut-off, submicron nickel particles were dispersed on the surface, vacuum sintered for 5 h at 800 °C, and then sputtered with TiN [159]. The nickel particles created a smoother surface with smaller pores, so a thinner palladium alloy layer

with fewer defects could be deposited. The TiN served as a diffusion barrier to prevent metallic interdiffusion [164]. A 3 μm thick Pd89–Ni11 film was co-deposited via vacuum electrodeposition so that a H_2/N_2 ideal separation factor of 4600 was maintained at 350 °C for 60 days. Nitrogen permeated through a similar membrane without the diffusion barrier after 50 days. The pores of PSS have also been filled and smoothed by alumina [226]. The palladium deposited by EP penetrated the porous alumina layer for increased adhesion.

EP seems to be a preferred deposition method [4, 73, 84, 182]. Very thin films of palladium alloyed with silver or copper have been deposited by EP onto porous glass and asymmetric ceramic membranes comprised of zirconia, titania, or γ-alumina top layers on α-alumina. For example, recent inroads have been made in accurately co-depositing Pd–Ag alloys from a single electroless bath [147]. Alternatively, palladium and then silver or copper have been deposited sequentially within porous tubular supports [72, 150, 228]. Depending on film thickness and treatment temperature, a homogenous alloy is formed during annealing. The results also demonstrated that the ability to obtain higher permselectivity for minimum palladium coating thickness depends primarily on several factors including deposition method, thorough surface activation, plating rate, metal deposition sequence [186], film thickness, the effect of incomplete coverage near seals [73, 84], air or oxygen treatment and, most critically, the surface roughness of the support [84].

Another factor that influences palladium film permeability is grain size. There is some evidence that hydrogen permeates faster through materials with fewer grain boundaries such as polycrystalline metals instead of nanocrystalline metals [78, 170, 229]. Without deliberate adjustment of deposition parameters, CVD, EP, and PVD all form films with nanocrystalline grains (<50 nm). Thus, the hydrogen permeability may be nonideal and deviate substantially from values for bulk palladium. Sizeable grain growth can take place under mild conditions (≈500 °C). Annealing can also promote structural changes such as recrystallization, stress relief, surface segregation, and atomic ordering that affect the hydrogen permeability and mechanical properties [108]. Depending on the temperature and the initial film configuration (thickness, type of support), annealing can result in densification or pinhole formation. At temperatures >600 °C, a decrease in hydrogen permeability is usually observed [146]. Absorption of hydrogen by palladium also creates strain in a membrane that can affect the permeability [109, 230–235]. Although usually more pronounced during membrane exposure to changing hydrogen concentration, the "uphill" component from non-Fickian stress-induced diffusion can affect the steady-state flux [109, 230, 231].

For a composite membrane that is relatively free of pinholes, there is apparently a limiting palladium thickness that depends primarily on support quality and somewhat on deposition method. For example, unmodified PSS requires at least 10 μm of palladium to bridge a majority of the pores [149, 227]. Invariably, a few macrodefects remain that preclude the attainment of perfect hydrogen permselectivity. On porous asymmetric α-alumina supports with 100–200 nm pore size on the surface, approximately 7 μm appears to be the limiting thickness that is proba-

bly needed to cover the largest pores and below which permselectivity falls [84, 180]. Thicker films are more thermally stable [149]. By optimizing the EP process, films fairly impervious to gases other than hydrogen and as thin as 1.2 μm have been prepared on asymmetric α-alumina supports with 200 nm pore size [73]. Supercritical electrodeposition has been used to deposit defect-free films as thin as ~2 μm onto porous ceramic supports [137]. Supercritical electrodeposition avoids the drawback of conventional electroplating–hydrogen embrittlement of the palladium deposit. Through careful control of the CVD conditions, including a pressure differential across the membrane during deposition, a porous (150 nm pore size) α-alumina tube was coated with 2–4 μm of palladium. Hydrogen fluxes of 0.1–0.2 mol were measured at 300 °C with a H_2/N_2 ideal separation factor >5000.

Thinner films can be deposited onto membrane supports with smaller pores, as in the case of porous α-alumina tubes coated with zirconia (50 nm pores), where Pd–Cu films of 1–4 μm thickness were sufficient to obtain high permselectivity (H_2/N_2 ≥7000) [68]. In another study, sol–gel γ-alumina with nominally 4 nm pores was coated onto α-alumina disks, and almost defect-free (H_2/He permselectivity ~4000) Pd82–Ag18 films as thin as 175 nm were deposited by magnetron sputtering [78]. The hydrogen flux measured through the membrane was low because, although a sweep gas was used, both sides of the membrane were kept at the same pressure (101 kPa). The aforementioned favorable results can be attributed to the narrow pore size distribution and low surface roughness of the support membranes. However, in many cases reproducibility is not readily achievable, suggesting that support quality may vary [68].

The economics of incorporating palladium membranes into an industrial process are modestly favorable. Cost estimations rely on the assumption that the membranes will have complete permselectivity for hydrogen and enable a membrane reactor to operate at reduced temperature [236]. For an overall membrane system, a major cost is the recompression of purified hydrogen, resulting in increased electrical power consumption [236]. Estimates for maximum acceptable palladium thickness vary from 5 to 20 μm [237, 238]. For palladium film thicknesses approaching 2 μm or less, the fraction of module cost from palladium shrinks to a small percentage of overall cost, with the membrane support being the bulk of the expense [148]. The possibility of reclaiming the palladium should also be considered.

3.9 Summary and Outlook

In this chapter, palladium-based hydrogen separating membranes were covered, numerous synthesis methodologies were explored, and some promising examples were given. Palladium alloy membranes are essential for hydrogen purification in some niche applications, although more widespread deployment remains elusive. Palladium alloy composite membranes that meet the criteria for industrial viability including high hydrogen permeance, reduced cost, robust operation at

high temperatures, and degradation resistance under gaseous contaminants are being developed methodically. From recent results, it is clear that breakthroughs have been made. Advances in the reproducible deposition of ultra-thin, pinhole-free, palladium-alloy films of a specific composition have been possible through the use of optimized deposition techniques coupled with the modification of porous supports to create smoother surfaces with well defined pore structure that may also serve as interlayer barriers that inhibit metallic interdiffusion.

Work continues with the hope that a sufficiently robust membrane module will be developed for industrial-scale hydrogen separations and hydrogen economy applications. However, key parameters limiting further progress remain, for example, the quality and cost of commercially available membranes as substrates. Integration of membranes into a process requires a systems approach. Membranes with hydrogen fluxes sufficient for most applications have been tested, but limitations still exist with regard to strength and the ability to thermally cycle without forming pinholes. Reproducible scale-up of membrane fabrication will be challenging. Continued innovation will naturally rely on the development of materials and fabrication methods to optimize membrane performance while minimizing membrane cost. It will be fascinating to see what mature forms the technology will encompass.

Acknowledgements

Many thanks to M. C. Timmers for editing. The author is also grateful for assistance from B.P. Nolen, M.R. Cortez, J.I. Abes and the Los Alamos National Laboratory Research Library for obtaining references.

References

1 *Materials Research for Separations Technologies: Energy and Emission Reduction Opportunities*, BCS, Inc. and Oak Ridge National Laboratory, 2005, p. 103.
2 J. Vaenselaar, Sustainable growth and chemical engineering, *Chem. Eng. Tech.* **2003**, *26*, 868–874.
3 D. zur Megede, Fuel processors for fuel cell vehicles, *J. Power Sources* **2002**, *106*, 35–41.
4 R. Hughes, Composite palladium membranes for catalytic membrane reactors, *Membr. Technol.* **2001**, *131*(3), 9–13.
5 T. Graham, Absorption and separation of gases by colloid septa, *Philos. Trans. R. Soc.* **1866**, *156*(399), 426–431.
6 W.O. Snelling, Apparatus for separating gases, U.S. Patent 1 174 631, Mar. 7, 1916.
7 R. C. Hurlbert, J. O. Konecny, Diffusion of hydrogen through palladium, *J. Chem. Phys.* **1961**, *34*(2), 655–658.
8 C. Lanzerstorfer, H. Puxbaum, The palladium/silver membrane focusing injector–a new inlet system for thermal desorption capillary gas chromatography, *Fresenius J. Anal. Chem.* **1991**, *339*(4), 223–225.
9 J. E. Lovelock, P. G. Simmonds, G. R. Shoemake, The palladium generator–separator–a combined electrolytic source and sink for hydrogen in closed

circuit gas chromatography, *Anal. Chem.* **1970**, *42*(9), 969–973.
10 P. G. Simmonds, G. R. Shoemake, J. E. Lovelock, Palladium-hydrogen system; efficient interface for gas chromatography-mass spectrometry, *Anal. Chem.* **1970**, *42*(8), 881–885.
11 O. M. Katz, E. A. Gulbransen, Permeability and diffusivity of hydrogen through a palladium tube, *Rev. Sci. Instrum.* **1960**, *31*(6), 615–617.
12 J. R. Young, Purity of hydrogen permeating through Pd, Pd-25% Ag, and Ni, *Rev. Sci. Instrum.* **1963**, *34*(8), 891–892.
13 F. J. Ackerman, G. J. Koskinas, A model for predicting the permeation of hydrogen–deuterium–inert gas mixtures through palladium tubes, *Ind. Eng. Chem. Fundam.* **1972**, *11*(3), 332–338.
14 Palladium curbs hydrogen waste, *The Iron Age*, **1964**, *193*(19), 86.
15 Palladium keeps hydrogen pure for bright annealing lines, *The Iron Age* **1963**, *191* (Feb 21), 140–141.
16 J. B. Hunter, A new hydrogen purification process, *Platinum Met. Rev.* **1960**, *4*(4), 130–131.
17 P. M. Roberts, D. A. Stiles, Palladium alloy diffusion units, *Platinum Met. Rev.* **1969**, *13*(4), 141.
18 E. J. Serfass, H. Silman, Hydrogen production and purification by the diffusion process, *The Chem. Eng.* **1965**, *192*, CE266–271.
19 A. S. Darling, The diffusion of hydrogen through palladium, *Platinum Met. Rev.* **1958**, *2*(1), 16–22.
20 A. J. DeRosset, Diffusion of hydrogen through palladium membranes, *Ind. Eng. Chem.* **1960**, *52*(6), 525–528.
21 H. Connor, Hydrogen diffusion through silver-palladium foil electrodes, *Platinum Met. Rev.*, **1966**, *10*(4), 127.
22 G. L. Matlack, A fully integrated hydrogen diffusion system, *Plat. Met. Rev.* **1969**, *13*(1), 26–27.
23 M. J. Cole, The generation of pure hydrogen for industrial applications, *Platinum Met. Rev.* **1985**, *25*(1), 12–13.
24 J. E. Philpott, Hydrogen diffusion technology: Commercial applications of Pd membranes, *Platinum Met. Rev.* **1985**, *29*(1), 12.
25 J. E. Philpott, The on-site production of hydrogen, *Platinum Met. Rev.* **1976**, *20*(4), 110–113.
26 J. Han, I.-S. Kim, K.-S. Choi, High purity hydrogen generator for on-site hydrogen production, *Int. J. Hydrogen Energy* **2002**, *27*, 1043–1047.
27 W. C. Pfefferle, Process for Dehydrogenation, U.S. Patent 3 290 406, Dec. 6, 1966.
28 V. M. Gryaznov, Membrane catalysts of the 21st century, *Petroleum Chem.* **1995**, *35*(3), 181–188.
29 J. Miyazaki, T. Kajiyama, K. Matsumoto, H. Fujiwara, M. Yatabe, Ultra high purity hydrogen gas supply system with liquid hydrogen, *Int. J. Hydrogen Energy* **1996**, *21*(5), 335–341.
30 Johnson-Matthey, *Pureguard™ palladium alloy diffusion cells*, 2005 (available at www.pureguard.net/html/products.html).
31 R. E. Buxbaum, 2005, www.rebresearch.com/products.html.
32 *Wah Chang Technical Data Sheet: Hydrogen Metal Membranes*, 2005 (available at www.wahchang.com/wahchang/pages/products/hmm/hmm.htm).
33 G. P. Glazunov, E. D. Volkov, A. Hassanein, Bimetallic diffusion membranes: Possible use for active hydrogen recycling control, in *Hydrogen & Helium Recycling at Plasma Facing Materials*, August 22–24, 2002, Argonne, Illinois, USA, Kluwer Academic Publishers, Norwell, MA, 2002.
34 N. Iniotakis, W. Frohling, C. B. Von Der Decken, Filter for the selective removal of tritium from the coolant gas of a high temperature nuclear reactor, *J. Less-Common Met.* **1983**, *89*, 465–474.
35 V. M. Gryaznov, Metal containing membranes for the production of ultrapure hydrogen and the recovery of hydrogen isotopes, *Sep. Purif. Meth.* **2000**, *29*(2), 171–187.
36 S. Konishi, M. Hara, K. Okuno, Plasma exhaust processing by palladium permeation and electrolytic oxidation and reduction, *Fusion Technol.* **1996**, *30*, 890–894.
37 D. L. Luo, Y. F. Xiong, J. F. Song, and G. Q. Huang, Hydrogen isotope separation factor measurement for single

stage hydrogen separators and parameters for a large-scale separation system, *Fusion Sci. Technol.* **2005**, *48*(1), 156–158.

38 V. V. Rumyantsev, V. M. Shatalov, G. Y. Misuna, Gas separation of hydrogen isotopes by means of multicell metal membrane, *Desalination* **2002**, *148*, 293–296.

39 S. Tosti, V. Violante, Numerical approach for a study of the hydrogen isotopes separation by palladium alloy membranes, *Fusion Eng. Des.* **1998**, *43*(1), 93–100.

40 K. Aoki, Y. Ogata, K. Kusakabe, S. Morooka, Applicability of palladium membrane for the separation of protium and deuterium, *Int. J. Hydrogen Energy* **1998**, *23*(5), 325–332.

41 V. V. Latyshev, V.M. Bystritskiy, Interaction of hydrogen isotopes and diffusion membranes made from palladium alloys, *Phys. Met. Metall.* **1991**, *71*(6), 1–20.

42 Y. I. Aristov, V. N. Parmon, G. Cacciola, N. Giordano, High-temperature chemical heat pump based on reversible catalytic reactions of cyclohexane-dehydrogenation/benzene-hydrogenation: Comparison of the potentialities of different flow diagrams, *Int. J. Energy Res.* **1993**, *17*, 293–303.

43 Y. M. Baranov, L. V. Stolyarova, M. F. Shopshin, T. K. Melik-Akhnazarov, T. S. Aleksashkina, Modern processes of hydrogen recovery from refinery gas, *Chem. Technol. Fuels Oils* **1992**, *27*(5), 285–287.

44 J. Shu, B. P. A. Grandjean, A. Van Neste, S. Kaliaguine, Catalytic palladium-based membrane reactors: A review, *Can. J. Chem. Eng.* **1991**, *69*(5), 1036–1060.

45 S. Uemiya, State-of-the-art of supported metal membranes for gas separation, *Sep. Purif. Methods* **1999**, *28*(1), 51–85.

46 S. Paglieri, J. D. Way, Innovations in palladium membrane research, *Sep. Purif. Methods* **2002**, *31*(1), 1–169.

47 J. Philpott, D. R. Coupland, Metal Membranes for Hydrogen Diffusion and Catalysis, in *Hydrogen Effects in Catalysis*, Z. Paál, P. G. Menon (Eds.), Marcel Dekker, Inc., New York, 1988, pp. 679–964.

48 G. J. Grashoff, C. E. Pilkington, C. W. Corti, The purification of hydrogen: A review of the technology emphasising the current status of palladium membrane diffusion, *Platinum Met. Rev.* **1983**, *27*, 157–169.

49 J. Ali, A. Baiker, Dehydrogenation of methylcyclohexane to toluene in a pilot-scale membrane reactor, *Appl. Catal. A* **1997**, *155*(1), 41–57.

50 V. M. Gryaznov, Conjugation of reactions using membrane catalysts, *Kinet. Catal.* **1971**, *12*(3), 564–568.

51 P. Ferreira-Aparicio, M. Benito, K. Kouachi, S. Menad, Catalysis in membrane reformers: A high-performance catalytic system for hydrogen production from methane, *J. Catal.* **2005**, *231*, 331–343.

52 H. Amandusson, L.-G. Ekedahl, H. Dannetun, Alcohol dehydrogenation over Pd versus PdAg membranes, *Appl. Catal.* **2001**, *217*, 157–164.

53 V. Gryaznov, A century of studying the mechanism of catalytic reactions of hydrogen redistribution, *Russ. J. Phys. Chem* **1999**, *73*(10), 1723–1726.

54 B. J. Wood, Dehydrogenation of cyclohexane on a hydrogen-porous membrane, *J. Catal.* **1968**, *11*, 30–34.

55 V. M. Gryaznov, A. P. Maganyuk, A. N. Karavanov, V. A. Naumov, Effect of phase composition of palladium–ruthenium membrane catalyst on selectivity of dehydrolinalool hydrogenation, *Bull. Acad. Sci. USSR* **1984**, *33*(4), 853–855.

56 A. Basile, L. Paturzo, A. Vazzana, Membrane reactor for the production of hydrogen and higher hydrocarbons from methane over Ru/Al_2O_3 catalyst, *Chem. Eng. J.* **2003**, *93*(1), 31–39.

57 S. Assambumrungrat, K. Suksomboon, P. Praserthdam, T. Tagawa, S. Goto, Simulation of a palladium membrane reactor for dehydrogenation of ethylbenzene, *J. Chem. Eng. Jpn.* **2002**, *35*(3), 263–273.

58 A. M. Adris, C. J. Lim, J. R. Grace, The fluidized bed membrane reactor for steam methane reforming: Model verification and parametric study, *Chem. Eng. Sci.* **1997**, *52*, 1609–1622.

59 J. Tong, Y. Matsumura, H. Suda, K. Haraya, Experimental study of steam reforming of methane in a thin (6 μm)

Pd-based membrane reactor, *Ind. Eng. Chem. Res.* **2005**, *44*(5), 1454–1465.

60 E. Johannessen, K. Jordal, Study of a H_2 separating membrane reactor for methane steam reforming at conditions relevant for power process with CO_2 capture, *Energy Conversion Management* **2005**, *46*(7–8), 1059–1071.

61 S. A. Birdsell, R.S. Willms, Tritium recovery from tritiated water with a two-stage palladium membrane reactor, *Fusion Eng. Des.* **1998**, *39–40*, 1041–1048.

62 K. L. Sessions, Processing tritiated water at the Savannah River Site: A production-scale demonstration of a palladium membrane reactor, *Fusion Sci. Technol.* **2005**, *48*(1), 91–96.

63 K. Munakata, B. Bornschein, D. Corneli, M. Glugla, Numerical simulation of membrane reactor for detritiation of plasma exhaust gas, *Fusion Sci. Technol.* **2005**, *48*(1), 17–22.

64 E. Kikuchi, Hydrogen-permselective membrane reactors, *CATTECH*, **1997**, *1*(1), 67–74.

65 G. Centi, R. Dittmeyer, S. Perathoner, M. Reif, Tubular inorganic catalytic membrane reactors: Advantages and performance in multiphase hydrogenation reactions, *Catal. Today* **2003**, *79–80*, 139–149.

66 J. N. Armor, Overcoming equilibrium limitations in chemical processes, *Appl. Catal.* **2001**, *222*, 91–99.

67 K. K. Sirkar, P. V. Shanbhag, A. S. Kovvali, Membrane in a reactor: A functional perspective, *Ind. Eng. Chem. Res.* **1999**, *38*(10), 3715–3737.

68 F. Roa, J. D. Way, Influence of alloy composition and membrane fabrication on the pressure dependence of the hydrogen flux of palladium-copper membranes, *Ind. Eng. Chem. Res.* **2003**, *42*(23), 5827–5835.

69 S. Aggarwal, A. P. Monga, S. R. Perusse, R. Ramesh, V. Ballarotto, E. D. Williams, B. R. Chalamala, Y. Wei, R. H. Reuss, Spontaneous ordering of oxide nanostructures, *Science* **2000**, *287*, 2235–2237.

70 F. Roa, J. D. Way, The effect of air exposure on palladium-copper composite membranes, *Appl. Surf. Sci.* **2005**, *240*, 85–104.

71 F. Bustamante, R. M. Enick, R. P. Killmeyer, B. H. Howard, K. S. Rothenberger, A. V. Cugini, B. D. Morreale, M. V. Ciocco, Uncatalyzed and wall-catalyzed forward water-gas shift reaction kinetics, *AIChE J.* **2005**, *51*(5), 1440–1454.

72 L. Yang, Z. Zhang, X. Gao, Y. Guo, B. Wang, O. Sakai, H. Sakai, T. Takahashi, Changes in hydrogen permeability and surface state of Pd-Ag/ceramic composite membranes after thermal treatment, *J. Membr. Sci.*, **2005**, *252*, 145–154.

73 J. N. Keuler, L. Lorenzen, S. Miachon, Preparing and testing Pd films of thickness 1-2 micrometer with high selectivity and high hydrogen permeance, *Sep. Sci. Technol.* **2002**, *37*(2), 379–401.

74 R. G. Musket, Effects of contamination on the interactions of hydrogen gas with palladium: A review, *J. Less-Common Met.* **1976**, *45*, 173–183.

75 A. P. Mishchenko, V. M. Gryaznov, M. E. Sarylova, Effect of thermodiffusion and chemical treatment of the surface of membrane catalysts made from palladium alloys with ruthenium and rhodium on penta-1,3-diene hydrogenation, *Bull. Acad. Sci. USSR* **1991**, *40*(6 (1)), 1154–1157.

76 A. B. Antoniazzi, A. A. Haasz, P. C. Stangeby, The effect of adsorbed carbon and sulphur on hydrogen permeation through palladium, *J. Nucl. Mater.* **1989**, *162–164*, 1065–1070.

77 J. P. Collins, R. W. Schwartz, R. Sehgal, T. L. Ward, C. J. Brinker, G. P. Hagen, C. A. Udovich, Catalytic dehydrogenation of propane in hydrogen permselective membrane reactors, *Ind. Eng. Chem. Res.* **1996**, *35*(12), 4398–4405.

78 B. A. McCool, Y.S. Lin, Nanostructured thin palladium-silver membranes: Effects of grain size on gas permeation properties, *J. Mater. Sci.* **2001**, *36*, 3221–3227.

79 N. R. Roshan, A. P. Mishchenko, V. P. Ployakova, N. I. Parfenova, E. M. Savitsky, E. A. Voitekhova, V. M. Gryaznov, M. E. Sarylova, The effect of the surface state on the hydrogen permeability and the catalytic activity of palla-

dium alloy membranes, *J. Less-Common Met.* **1983**, *89*, 423–428.
80 A. F. Y. Al-Shammary, I. T. Caga, J. M. Winterbottom, A. Y. Tata, I. R. Harris, Palladium-based diffusion membranes as catalysts in ethylene hydrogenation, *J. Chem. Tech. Biotechnol.* **1991**, *52*, 571–585.
81 M. C. Iliuta, B. P. A. Grandjean, F. Larachi, Methane nonoxidative aromatization over Ru–Mo/HZSM-5 at temperatures up to 973 K in a palladium–silver/stainless steel membrane reactor, *Ind. Eng. Chem. Res.* **2003**, *42*, 323–330.
82 H. Amandusson, L.-G. Ekedahl, H. Dannetun, The effect of CO and O_2 on hydrogen permeation through a palladium membrane, *Appl. Surf. Sci.* **2000**, *153*, 259–267.
83 A. Kulprathipanja, G. O. Alptekin, J. L. Falconer, J. D. Way, Effects of water gas shift gases on Pd-Cu alloy membrane surface morphology and separation properties, *Ind. Eng. Chem. Res.* **2004**, *43*(15), 4188–4198.
84 V. Höllein, M. Thornton, P. Quicker, R. Dittmeyer, Preparation and characterization of palladium composite membranes for hydrogen removal in hydrocarbon dehydrogenation reactors, *Catal. Today* **2001**, *67*, 33–42.
85 D.-W. Lee, Y.-G. Lee, S.-E. Nam, S.-K. Ihm, K.-H. Lee, Study on the variation of morphology and separation behavior of the stainless steel supported membranes at high temperature, *J. Membr. Sci.* **2003**, *220*, 137–153.
86 V. V. Latyshev, V. G. Gur'yanov, M. Y. Asliddinova, I. I. Cholovyan, G. I. Mironova, Hydrogen-stimulated penetration of impurities through palladium alloy membranes, *Russ. J. Phys. Chem.* **1991**, *65*(5), 755–757.
87 T. H. Hsiung, D. D. Christman, E. J. Hunter, A. R. Homyak, Methane formation on H_2 purification using a commercial Pd-Ag membrane, *AIChE J.* **1999**, *45*(1), 204–208.
88 I. G. Batirev, A. N. Karavanov, J. A. Leiro, M. Heinonen, J. Juhanoja, X-ray photoelectron and Auger electron spectroscopies and surface segregations in Pd-Rh and Pd-Ru membrane catalysts, *Surf. Int. Anal.* **1995**, *23*(1), 50–54.
89 J. Shu, B. E. W. Bongondo, B. P. A. Grandjean, A. Adnot, S. Kaliaguine, Surface segregation of Pd-Ag membranes upon hydrogen permeation, *Surf. Sci.* **1993**, *291*, 129–138.
90 Y. Cao, H. Li, J. A. Szpunar, W. T. Shmayda, Modeling of hydrogen trapping in the deformed Pd and Pd77Ag23 alloy, *Mater. Sci. Eng.* **2004**, *379*, 173–180.
91 E. S. Carnell, S. P. Wach, Transport coefficients and energetics of hydrogen in palladium, *Surf. Coatings Technol.* **1986**, *28*, 339–345.
92 M. Amano, C. Nishimura, M. Komaki, Effects of high concentration CO and CO_2 on hydrogen permeation through the palladium membrane, *Mater. Trans., JIM*, **1990**, 31(5), 404–408.
93 D. J. Edlund, W. A. Pledger, Thermolysis of hydrogen sulfide in a metal-membrane reactor, *J. Membr. Sci.* **1993**, *77*, 255–264.
94 D. J. Edlund, W. A. Pledger, Catalytic platinum-based membrane reactor for removal of H_2S from natural gas streams, *J. Membr. Sci.* **1994**, *94*, 111–119.
95 M. Kajiwara, S. Uemiya, T. Kojima, Stability and hydrogen permeation behavior of supported platinum membranes in presence of hydrogen sulfide, *Int. J. Hydrogen Energy* **1999**, *24*(9), 839–844.
96 P. Kamakoti, B. D. Morreale, M. V. Ciocco, B. H. Howard, R. P. Killmeyer, A. V. Cugini, D. S. Sholl, Prediction of hydrogen flux through sulfur-tolerant binary alloy membranes, *Science* **2005**, *307*, 569–573.
97 K. J. Bryden, J. Y. Ying, Nanostructured palladium-iron membranes for hydrogen separation and membrane hydrogenation reactions, *J. Membr. Sci.* **2002**, *203*, 29–42.
98 A. Kulprathipanja, G. O. Alptekin, J. L. Falconer, J. D. Way, Pd and Pd-Cu membranes: Inhibition of H_2 permeation by H_2S, *J. Membr. Sci.* **2005**, *254*, 49–62.
99 J. K. Ali, E. J. Newson, D. W. T. Rippin, Deactivation and regeneration of Pd-Ag membranes for dehydrogenation reactions, *J. Membr. Sci.* **1994**, *89*, 171–184.
100 A. Li, R. Hughes, The effect of carbon monoxide and steam on the hydrogen

permeability of a Pd/stainless steel membrane, *J. Membr. Sci.* **2000**, *165*, 135–141.

101 Y. Sakamoto, F. L. Chen, Y. Kinari, F. Sakamoto, Effect of carbon monoxide on hydrogen permeation in some palladium-based alloy membranes, *Int. J. Hydrogen Energy* **1996**, *21*(11/12), 1017–1024.

102 E. Xue, M. O'Keeffe, J. R. H. Ross, Water-gas shift conversion using a feed with a low steam to carbon monoxide ratio and containing sulphur, *Catal. Today* **1996**, *30*, 107–118.

103 S. Hara, K. Sakaki, N. Itoh, Decline in hydrogen permeation due to concentration polarization and CO hindrance in a palladium membrane reactor, *Ind. Eng. Chem. Res.* **1999**, *38*(12), 4913–4918.

104 D. Wang, T.B. Flannagan, K. L. Shananhan, Permeation of hydrogen through pre-oxidized Pd membranes in the presence and absence of CO, *J. Alloys Compd.* **2004**, *372*, 158–164.

105 H. Amandusson, L.-G. Ekedahl, H. Dannetun, Hydrogen permeation through surface modified Pd and PdAg membranes, *J. Membr. Sci.* **2001**, *193*, 35–47.

106 J. K. Ali, P. Hasler, E. J. Newson, D. W. T. Rippin, Irreversible poisoning of Pd-Ag membranes, *Int. J. Hydrogen Energy* **1994**, *19*(11), 877–880.

107 A. G. Knapton, Palladium alloys for hydrogen diffusion membranes: A review of high permeability materials, *Platinum Met. Rev.* **1977**, *21*(2), 44.

108 C. Anderton, N. Strother, J. Pote, R. Foley, K. Rebeiz, S. Nesbit, A. Craft, Influence of partial hydride formation/decomposition on the mechanical properties of palladium, *Scr. Mater.* **1996**, *35*(8), 1013–1018.

109 A. Adrover, M. Giona, L. Capobianco, V. Violante, Effects of self-stress on hydrogen diffusion in Pd membranes in the coexistence of alpha and beta phases, *J. Alloys Compd.* **2004**, *368*, 287–297.

110 F. A. Lewis, *The Palladium Hydrogen System*, Academic Press, London, 1967.

111 B. A. Wilhite, M. A. Schmidt, K. F. Jensen, Palladium-based micromembranes for hydrogen separation: Device performance and chemical stability, *Ind. Eng. Chem. Res* **2004**, *43*, 7083–7091.

112 L. C. W. Hobbis, E. R. Harrison, Use of nickel diffusion tubes for the purification of hydrogen, *Rev. Sci. Instrum.* **1956**, *27*(4), 238.

113 D. Slovetsky, Russian membrane technology produces hydrogen, *Membr. Technol.* **2002**, *142*(2), 1.

114 V. I. Koshel', V. V. Latyshev, V. A. Gol'tsov, Thermal concentration dilation of V-2 membrane alloy in a hydrogen atmosphere, *Sov. Mater. Sci.* **1984**, *20*(6), 551–553.

115 F. N. Berseneva, I. A. Mikhaylova, N. I. Timofeyev, The influence of hydrogen on the phase composition and physicomechanical properties of the membrane palladium-based alloy V-1, *Phys. Met. Metall.* **1992**, *72*(2), 151–156.

116 I. Y. Machikina, G. A. Barannikova, V. A. Gol'tsov, V. V. Latyshev, S. K. Kurumchina, N. I. Timofeev, Mechanical and diffusion characteristics of palladium-base type V-3 phase-hardened alloys for hydrogen diffusion filters, *Sov. Mater. Sci.* **1985**, *21*(1), 97–99.

117 D. Wang, T. Flanagan, K. L. Shanahan, Hydrogen permeation measurements of partially internally oxidized Pd-Al alloys in the presence and absence of CO, *J. Membr. Sci.* **2005**, *253*, 165–173.

118 D. W. Kim, K. Y. Um, H. G. Kim, I. S. Lee, S. H. Kim, J. S. Park, S. J. Ryi, S. H. Cho, A Pd-Cu-Ni ternary alloyed membrane on porous nickel support prepared by sputtering and copper reflow, *Jpn. J. Appl. Phys.* **2005**, *744*(7), L233–L234.

119 I. N. Avertseva, M. V. Raevskaya, Hydrogen permeability and mechanical properties of palladium alloyed with ruthenium, samarium, yttrium, and neodymium, *Phys. Met. Metall.* **1994**, *77*(5), 518–521.

120 N. I. Timofeev, F. N. Berseneva, V. M. Makarov, New palladium-based membrane alloys for separation of gas mixtures to generate ultrapure hydrogen, *Int. J. Hydrogen Energy* **1994**, *19*(11), 895–898.

121 K. J. Bryden, J. Y. Ying, Synthesis and structural characteristics of mechnically alloyed nanostructured palladium-

yttrium, *Mater. Sci. Forum* **1996**, *225–227*(2), 895–900.

122 H. Gao, Y. S. Lin, Y. Li, B. Zhang, Chemical Stability and its improvement of palladium-based metallic membranes, *Ind. Eng. Chem. Res* **2004**, *43*(22), 6920–6930.

123 G. A. Kok, A. Noordermeer, B. E. Nieuwenhuys, Effect of alloying on the adsorption of CO on palladium; a comparison of the behaviour of PdAg(111), PdCu(111) and Pd(111) surfaces, *Surf. Sci*. **1985**, *152/153*, 505–512.

124 A. Noordermeer, G. A. Kok, B. E. Nieuwenhuys, A comparative study of the behaviour of the PdAg(111) and Pd(111) surfaces towards the interaction with hydrogen and carbon monoxide, *Surf. Sci*. **1986**, *165*, 375–392.

125 A. Noordermeer, G. A. Kok, B. E. Nieuwenhuys, Comparison between the adsorption properties of Pd(111) and PdCu(111) surfaces for carbon monoxide and hydrogen, *Surf. Sci*. **1986**, *172*, 349–362.

126 A. S. Zetkin, G. E. Kagan, A. N. Varaksin, E. S. Levin, Diffusion and penetrability of deuterium in the alloy Pd-53 at. Cu, Sov. *Phys. Solid State*, **1992**, *34*(1), 83–85.

127 D. L. McKinley, Metal alloy for hydrogen separation and purification, U.S. Patent 3 350 845, Nov. 7, 1967.

128 D. L. McKinley, Method for hydrogen separation and purification, U.S. Patent 3 439 474, Apr. 22, 1969.

129 D. Edlund, A membrane reactor for H_2S decomposition, in *Proceedings of the Advanced Coal-Fired Power Systems '96 Review Meeting*, Jul. 16–18, 1996, Morgantown, West Virginia, USA.

130 H. T. Hoang, H. D. Tong, F. C. Gielens, H. V. Jansen, M. C. Elwenspoek, Fabrication and characterization of dual sputtered Pd-Cu alloy films for hydrogen separation membranes, *Mater. Lett*. **2004**, *58*, 525–528.

131 P. H. Thornton, Palladium and palladium/copper thin flat membranes, U.S. Patent 6 171 712, Jan. 9, 2001.

132 J. Piper, Diffusion of hydrogen in copper–palladium alloys, *J. Appl. Phys*. **1966**, *37*(2), 715–721.

133 P. R. Subramanian, D. J. Chakrabarti, D. E. Laughlin (Eds.), *Phase Diagrams of Binary Copper Alloys. Monograph Series on Alloy Phase Diagrams*, The Materials Information Society, Materials Park, OH, 1994, pp. 308–318.

134 B. D. Morreale, M. V. Ciocco, B. H. Howard, R. Killmeyer, A. V. Cugini, R. M. Enick, Effect of hydrogen-sulfide on the hydrogen permeance of palladium-copper alloys at elevated temperatures, *J. Membr. Sci*. **2004**, *241*, 219–224.

135 F. Roa, M. J. Block, J. D. Way, The influence of alloy composition on the H_2 flux of composite Pd–Cu membranes, *Desalination* **2002**, *147*(1-3), 411–416.

136 N. E. Fernandes, S. M. Fisher, J. C. Poshusta, D. G. Vlachos, M. Tsapatsis, J. J. Watkins, Reactive deposition of metal thin films within porous supports from supercritical fluids, *Chem. Mater.* **2001**, *13*(6), 2023.

137 S. Uemiya, E. Makino, I. Miyazaki, H. Kawamura, R. Yoshiie, H. Moritomi, M. Nishimura, Fabrication of a palladium composite membrane using supercritical wet plating technique in *Proceedings of the 8th International Conference on Inorganic Membranes*, Jul. 18–22, 2004, Cincinnati, OH, Adams Press, Chicago, IL.

138 Z. Y. Li, H. Naeda, K. Kusakabe, S. Morooka, H. Anzai, S. Akiyama, Preparation of palladium-silver alloy membranes for hydrogen separation by the spray pyrolysis method, *J. Membr. Sci*. **1993**, *78*, 247–254.

139 J. Kim, Y.S. Lin, Palladium-modified yttria-stablilized zirconia membranes, *Ind. Eng. Chem. Res*. **2000**, *39*(6), 2124–2126.

140 R. M. de Vos, H. Vereij, High-selectivity, high-flux silica membranes for gas separation, *Science*, **1998**, *279*(13), 1710–1711.

141 N. Itoh, T. Akiha, T. Sato, Preparation of thin palladium composite membrane tube by a CVD technique and its hydrogen permselectivity, *Catal. Today* **2005**, *104*, 231–237.

142 G. Y. Meng, L. Huang, M. Pan, C. S. Chen, D. K. Peng, Preparation and characterization of Pd and Pd–Ni alloy membranes on porous substrates by

MOCVD with mixed metal β-diketone precursors, *Mater. Res. Bull.*, **1997**, *32*(4), 385–395.

143 G. Xomeritakis, Y.-S. Lin, CVD synthesis and gas permeation properties of thin palladium/alumina membranes, *AIChE J.* **1998**, *44*(1), 174–183.

144 G. O. Mallory, J. B. Hajdu (Eds.), *Electroless Plating*, American Electroplaters & Surface Finishers Society, New York, Noyes Publications, Orlando, FL, 1990, p. 539.

145 P. D. Cobden, V. V. Gorodetskii, B. E. Nieuwenhuys, Field emission microscope study of the initial behaviour of the palladium-hydrogen system at low temperatures, *Surf. Sci.* **1999**, *432*, 61–68.

146 S. N. Paglieri, K. Y. Foo, J. D. Way, J. P. Collins, D. L. Harper-Nixon, A new preparation technique for Pd/alumina membranes with enhanced high temperature stability, *Ind. Eng. Chem. Res.* **1999**, *38*(5), 1925–1936.

147 D. A. Pacheco Tanaka, M. A. Llosa Tanco, S. Niwa, Y. Wakui, F. Mizukami, T. Namba, T. M. Suzuki, Preparation of palladium and silver alloy membrane on a porous alpha alumina tube via simultaneous electroless plating, *J. Membr. Sci.* **2005**, *247*, 21–27.

148 K. Kuraoka, H. Zhao, T. Yazawa, Pore-filled palladium-glass composite membranes for hydrogen separation by novel electroless plating technique, *J. Mater. Sci.* **2004**, *39*, 1879–1881.

149 C. Su, T. Jin, K. Kuraoka, Y. Matsumura, T. Yazawa, Thin palladium film supported on SiO_2-modified porous stainless steel for a high-hydrogen-flux membrane, *Ind. Eng. Chem. Res.* **2005**, *44*(9), 3053–3058.

150 F. Roa, J. D. Way, R. L. McCormick, S. N. Paglieri, Preparation and characterization of Pd-Cu composite membranes for hydrogen separation, *Chem. Eng. J.* **2003**, *93*(1), 11–22.

151 J. Shu, B. P. A. Grandjean, S. Kaliaguine, Morphological study of hydrogen permeable Pd membranes, *Thin Solid Films* **1994**, *252*(1), 26–31.

152 S. Uemiya, T. Matsuda, E. Kikuchi, Hydrogen permeable palladium-silver alloy membrane supported on porous ceramics, *J. Membr. Sci.* **1991**, *56*, 315–325.

153 S. Uemiya, N. Sato, H. Ando, Y. Kude, T. Matsuda, E. Kikuchi, Separation of hydrogen through palladium thin film supported on a porous glass tube, *J. Membr. Sci.* **1991**, *56*, 303–313.

154 W. H. Lin, H.F. Chang, Characterizations of Pd-Ag membrane prepared by sequential electroless deposition, *Surf. Coatings Technol.* **2005**, *194*, 157–166.

155 H. I. Chen, C. Y. Chu, T. C. Huang, Characterization of Pd/Al_2O_3 composite membrane by electroless codeposition, *Thin Solid Films* **2004**, *460*, 62–71.

156 Y. H. Ma, I. P. Mardilovich, E. E. Engwall, Thin composite palladium and palladium/alloy membranes for hydrogen separation, *Ann. N.Y. Acad. Sci.* **2003**, *984*, 346–360.

157 T.-C. Huang, M.-C. Wei, H.-I. Chen, Preparation of hydrogen-permselective palladium-silver alloy composite membranes by electroless co-deposition, *Sep. Purif. Technol.* **2003**, *32*, 239–245.

158 J. Shu, B. P. A. Grandjean, E. Ghali, S. Kaliaguine, Simultaneous deposition of Pd and Ag on porous stainless steel by electroless plating, *J. Membr. Sci.* **1993**, *77*, 181–195.

159 S. E. Nam, K.H. Lee, Preparation and characterization of palladium alloy composite membranes with a diffusion barrier for hydrogen separation, *Ind. Eng. Chem. Res.* **2005**, *44*, 100–105.

160 T. L. Ward, S. B. Rathod, A. S. Chellappa, T. R. Vencill, Fabrication of thin supported palladium alloy membranes for fuel cell applications, in *Proceedings of the 8th International Conference on Inorganic Membranes*, Jul. 18–22, 2004, Cincinnati, OH, Adams Press, Chicago, IL.

161 H. D. Tong, A. H. J. vanden Berg, J. G. E. Gardeniers, H. V. Jansen, F. C. Gielens, M. C. Elwenspoek, Preparation of palladium-silver alloy films by a dual-sputtering technique and its application in hydrogen separation membrane, *Thin Solid Films* **2005**, *479*, 89–94.

162 B. McCool, G. Xomeritakis, Y. S. Lin, Composition control and hydrogen permeation characteristics of sputter

deposited palladium-silver membranes, *J. Membr. Sci.* **1999**, *161*(1–2), 67–76.
163 H. J. Frost, Microstructural Evolution in Thin Films, *Mater. Char.* **1994**, *32*, 257–273.
164 J. Shu, A. Adnot, B. P. A. Grandjean, S. Kaliaguine, Structurally stable composite Pd-Ag alloy membranes: Introduction of a diffusion barrier, *Thin Solid Films* **1996**, *286*(1–2), 72–79.
165 S. V. Karnik, M. K. Hatalis, M. V. Kothare, Towards a palladium micromembrane for the water gas shift reaction: Microfabrication approach and hydrogen purification results, *J. Microelectromech. Syst.* **2003**, *12*(1), 93–100.
166 T. Akamatsu, K. Kume, K. Komiya, H. Yukawa, M. Morinaga, S. Yamaguchi, Electrochemical method for measuring hydrogen permeability through metals, *J. Alloys Compd.* **2005**, *393*, 302–306.
167 T. L. Ward, T. Dao, Model of hydrogen permeation behavior in palladium membranes. *J. Membr. Sci.* **1999**, *153*, 211–231.
168 P. L. Andrew, A. A. Haasz, Models for hydrogen permeation in metals, *J. Appl. Phys.* **1992**, *72*(7), 2749–2757.
169 A. D. Le Claire, Permeation of gases through solids, *Diffus. Defect Data* **1983**, *33*, 1–66.
170 S. A. Koffler, J. B. Hudson, G. S. Ansell, Hydrogen permeation through alpha-palladium, *Trans. Metall. Soc. AIME* **1969**, *245*, 1735–1740.
171 J. P. Collins, J. D. Way, Preparation and characterization of a composite palladium-ceramic membrane, *Ind. Eng. Chem. Res.* **1993**, *32*(12), 3006–3013.
172 T. N. Kompaniets, A.A. Kurdyumov, Surface processes in hydrogen permeation through metal membranes, *Prog. Surf. Sci.* **1984**, *17*, 75–152.
173 I. P. Mardilovich, E. Engwall, Y. H. Ma, Dependence of hydrogen flux on the pore size and plating surface topology of asymmetric Pd-porous stainless steel membranes, *Desalination* **2002**, *144*(1–3), 85–89.
174 Y. K. Baichtok, Y. A. Sokolinskii, M.B. Aizenbud, Limiting stage in the permeation of hydrogen through palladium alloy membranes, *Russ. J. Phys. Chem.* **1976**, *50*(6), 929–930.
175 P. A. Lessing, H. C. Wood, L. D. Zuck, Low temperature hydrogen transport using a palladium/copper membrane, *J. Mater. Sci.* **2003**, *38*, 2401–2408.
176 S.-E. Nam, K.-H. Lee, A study on the palladium-nickel composite membrane by vacuum electrodeposition, *J. Membr. Sci.* **2000**, *170*(1), 91–99.
177 R. Dittmeyer, V. Hollein, P. Quicker, P. Emig, G. Hausinger, F. Schmidt, Factors controlling the performance of catalytic dehydrogenation of ethylbenzene in palladium composite membrane reactors, *Chem. Eng. Sci.* **1999**, *54*(10), 1431–1439.
178 O. Schramm, A. Seidel-Morgenstern, Comparing porous and dense membranes for the application in membrane reactors, *Chem. Eng. Sci.* **1999**, *54*(10), 1447–1453.
179 K. S. Rothenberger, A. V. Cugini, B. H. Howard, R. P. Killmeyer, M. V. Ciocco, B. D. Morreale, R. M. Enick, F. Bustamante, I. P. Mardilovich, Y. H. Ma, High pressure hydrogen permeance of porous stainless steel coated with a thin palladium film via electroless plating, *J. Membr. Sci.* **2004**, *244*, 55–68.
180 Roa, F., J. D. Way, S. DeVoss, G. O. Alptekin, Effect of CO_2, CO, H_2S on a Pd-Cu ceramic composite membrane made by electroless plating in *Proceedings of the 8th International Conference on Inorganic Membranes*, Jul. 18–22, 2004, Cincinnati, OH, Adams Press, Chicago, IL.
181 E. Gobina, K. Hou, R. Hughes, Ethane dehydrogenation in a catalytic membrane reactor coupled with a reactive sweep gas, *Chem. Eng. Sci.* **1995**, *50*(14), 2311–2319.
182 P. P. A. C. Pex, Y. C. van Delft, L. A. Correia, H. M. van Veen, D. Jansen, J. W. Dijkstra, Palladium alloy membranes for energy efficient membrane reactors in *Proceedings of the 8th International Conference on Inorganic Membranes*, Jul. 18–22, 2004, Cincinnati, OH, Adams Press, Chicago, IL.
183 R. B. Belser, R. B., Alloying behavior of thin bimetal films, simultaneously or successively deposited, *J. Appl. Phys.* **1960**, *31*(3), 562–570.

184 X. L. Pan, N. Stroh, H. Brunner, G. X. Xiong, S. S. Sheng, Pd/ceramic hollow fibers for H_2 separation, *Sep. Purif. Technol.* **2003**, *32*, 265–270.

185 L. C. Witjens, J. H. Bitter, A. J. Van Dillen, W. M. Arnoldbik, F. H. P. M. Habraken, K. P. de Jong, Improving the control of the electroless plating synthesis of Pd/Ag membranes for hydrogen separation using Rutherford backscattering, *J. Membr. Sci.* **2005**, *254*, 241–248.

186 J. N. Keuler, L. Lorenzen, R. D. Sanderson, V. Prozesky, W. J. Przybylowicz, Characterising palladium-silver and palladium-nickel alloy membranes using SEM, XRD and PIXE, *Nucl. Instrum. Methods Phys. Res. B* **1999**, *158*, 678–682.

187 K.-N.P. Kumar, K. Keizer, A. J. Burggraaf, Stabilization of the porous texture of nanostructured titania by avoiding a phase transformation, *J. Mater. Sci. Lett.* **1994**, *13*, 59–61.

188 J. F. Silvain, O. Fouassier, Atomic force microscopy and x-ray photoelectron spectroscopy investigations of the morphology and chemistry of a $PdCl_2$/$SnCl_2$ electroless plating catalysis system adsorbed onto shape memory alloy particles. *J. App. Phys.* **2004**, *96*(9), 4945–4951.

189 J. Horkans, J. Kim, C. McGrath, L. T. Romankiw, A TEM study of the effect of accelerators on Pd-Sn colloidal catalysts and on the initiation of electroless Cu deposition on epoxy, *J. Electrochem. Soc.* **1987**, *134*(2), 300–304.

190 J. Luyten, J. Cooymans, W. Adriansens, R. Leysen, Alternative process routes for better ceramic membrane supports, *Ind. Ceram.* **2000**, *20*(1), 29–31.

191 V. M. Gryaznov, O. S. Serebryannikova, Y. M. Serov, M. M. Ermilova, A. N. Karavanov, A. P. Mischenko, N. V. Orekhova, Preparation and catalysis over palladium composite membranes, *Appl. Catal. A* **1993**, *96*(1), 15.

192 D. J. Edlund, J. McCarthy, The relationship between intermetallic diffusion and flux decline in composite-metal membranes: Implications for achieving long membrane lifetime, *J. Membr. Sci.* **1995**, *107*, 147–153.

193 N. M. Peachey, R. C. Snow, R. C. Dye, Composite Pd/Ta metal membranes for hydrogen separation, *J. Membr. Sci.* **1996**, *111*, 123–133.

194 S. Uemiya, Y. Kude, K. Sugino, N. Sato, T. Matsuda, E. Kikuchi, A palladium/porous-glass composite membrane for hydrogen separation, *Chem. Lett.* **1988**, 1687–1690.

195 P. P. Mardilovich, P. V. Kurman, A. N. Govyadinov, I. P. Mardilovich, R. Paterson, V. M. Gryaznov, Gas Permeability of Anodized Alumina Membranes with a Palladium–Ruthenium Alloy Layer, Russ. *J. Phys. Chem.* **1996**, *70*(3), 514–517.

196 R. S. Souleimanova, A.S. Mukasyan, A. Varma, Pd membranes formed by electroless plating with osmosis: H_2 permeation studies, *AIChE J.* **2002**, *48*(2), 262–268.

197 K. L. Yeung, A. Varma, Novel preparation techniques for thin metal-ceramic composite membranes, *AIChE J.* **1995**, *41(9)*, 2131.

198 S.-E. Nam, S.-H. Lee, K.H. Lee, Preparation of a palladium alloy composite membrane supported in a porous stainless steel by vacuum electrodeposition, *J. Membr. Sci.* **1999**, *153*, 163–173.

199 S. Morooka, S. Yan, S. Yokoyama, K. Kusakabe, Palladium membrane formed in macropores of support tube by chemical vapor deposition with cross-flow through a porous wall, *Sep. Sci. Technol.* **1995**, *30*(14), 2877–2889.

200 V. V. Latyshev, V. G. Gur'yanov, M. Y. Asliddinova, I. I. Cholovyan, G. I. Mironova, Hydrogen-stimulated impurity diffusion through palladium-alloy membranes, *Sov. Phys. Technol. Phys.* **1991**, *36*(9), 1058–1059.

201 K. S. Weil, J. T. Darsell, J. S. Hardy, J. Y. Kim, Advances in air brazing for use in joining and sealing high-temperature gas separation membranes in *Proceedings of the 18th Fossil Energy Materials Conference*, Jun. 2–4, 2004, Knoxville, TN, Oak Ridge National Laboratory.

202 S. Tosti, L. Bettinali, Diffusion bonding of Pd-Ag rolled membranes, *J. Mater. Sci.* **2004**, *39*, 3041–3046.

203 S. Tosti, A. Adrover, A. Basile, V. Camilli, G. Chiappetta, V. Violante, Characterization of thin wall Pd-Ag rolled mem-

branes, *Int. J. Hydrogen Energy* **2003**, *28*, 105–112.

204 H. Klette, Y. Larring, H. Raeder, R. Bredesen, A new method for preparation of Pd-based composite membranes in *Proceedings of the 8th International Conference on Inorganic Membranes*, Jul. 18–22, 2004, Cincinnati, OH, Adams Press, Chicago, IL.

205 W. Juda, C. W. Krueger, R. T. Bombard, Hydrogen Generators, WO Patent 00/05169, Feb. 3, 2000.

206 Y. Guo, G. Lu, X. Mo, Y. Wang, Vaporphase dehydrogenation of methanol to methyl formate in catalytic membrane reactor with Pd/SiO_2/ ceramic composite membrane, *Chem. Lett.* **2004**, *33*(12), 1628–1629.

207 F. Morón, M. P. Pina, E. Urriolabeitia, M. Menéndez, J. Santamaria, Preparation and characterization of Pd-zeolite composite membranes for hydrogen separation, *Desalination* **2002**, *147*(1–3), 425–431.

208 M. Kajiwara, S. Uemiya, T. Kojima, E. Kikuchi, Rhodium- and iridium-dispersed porous alumina membranes and their hydrogen permeation properties, *Catal. Today* **2000**, *56*(1–3), 83–87.

209 J. H. So, K. Y. Yoon, S. M. Yang, Preparations of metal impregnated porous inorganic membranes for hydrogen separation by multi-step pore modifications, *Korean J. Chem. Eng.* **1999**, *16*(2), 180–186.

210 M. Chai, Y. Yamashita, M. Machida, K. Eguchi, H. Arai, Preparation and characterization of metal-dispersed alumina membranes for selective separation of hydrogen, *J. Membr. Sci.* **1994**, *97*, 199–207.

211 H. Suda, K. Haraya. Gas transport properties of carbon molecular sieve membranes doped with nano particles in *Proceedings of the 8th International Conference on Inorganic Membranes,*, Jul. 18–22, 2004, Cincinnati, OH, Adams Press, Chicago, IL.

212 H. B. Zhao, G. X. Xiong, J. H. Gu, S. S. Sheng, H. Bauser, N. Stroh, K. Pflanz, Preparation and characterization of novel porous metal/ceramic catalytic membrane materials, *Catal. Today* **1995**, *25*(3-4), 237–240.

213 A. L. Athayde, R. W. Baker, P. Nguyen, Metal composite membranes for hydrogen separation, *J. Membr. Sci.* **1994**, *94*, 299–311.

214 J. H. Shim, S. M. Song, W. K. Her, I. G. Koo, W. M. Lee, Electrochemical Acceleration of hydrogen transfer through a methanol impermeable metallic barrier, *J. Electrochem. Soc.* **2003**, *150*, A1583–A1588.

215 T. Hejze, B. R. Gollas, R. K. Sauerbrey, M. Schmied, F. Hofer, J. O. Besenhard, Preparation of Pd-coated polymer electrolyte membranes and their application in direct methanol fuel cells, *J. Power Sources* **2005**, *140*, 21–27.

216 H. P. Hsieh, General characteristics of inorganic membranes, in *Inorganic Membranes Characterization and Applications*, R.R. Bhave (Ed.), van Nostrand Reinhold, New York, 1991.

217 X. Pan, M. Kilgus, A. Goldbach, Low-temperature H_2 and N_2 transport through thin $Pd_{66}Cu_{34}H_x$ layers, *Catal. Today* **2005**, *104*, 225–230.

218 R. L. Goldsmith, Membrane devices for pervaporation, gas separations, and perstraction with permeate and sweep conduit and method of use, U. S. Patent Application 2005/0092683, May 5, 2005.

219 F. C. Gielens, H. D. Tong, C. J. M. Van Rijn, M. A. G. Vorstman, J. T. F. Keurentjes, Microsystem technology for high-flux hydrogen separation membranes, *J. Membr. Sci.* **2004**, *243*, 203–213.

220 J. D. Holladay, Y. Wang, E. Jones, Review of developments in portable hydrogen production using microreactor technology, *Chem. Rev.* **2004**, *104*, 4767–4790.

221 M. V. Kothare, A. V. Pattekar, K. A. Alfadhel, L. Bleris, S. Mukherjee, Microreactors for efficient on-chip fuel processing and hydrogen generation, *Proc. SPIE* **2005**, *5592*, 241–253.

222 A. J. Franz, K. F. Jensen, M. A. Schmidt, S. Firebaugh, Integrated palladium-based micromembranes for hydrogen separation and hydrogenation/dehydrogenation reactions, U. S. Patent 6 541 676, Apr. 1, 2003.

223 H. Klette, R. Bredesen, Sputtering of very thin palladium-alloy hydrogen sep-

aration membranes, *Membr. Technol.* **2005**, *5*, 7–9.

224 B. Lanning, J. Arps, Cost-effective method for producing self supported palladium alloy membranes for use in efficient production of coal derived hydrogen, *Quarterly Technical Progress Report*, Apr. 30, 2005, DOE report #DE00840474 (available at www.osti.gov/bridge/servlets/purl/840474-A3mSZG/native/840474.pdf).

225 N. Jemaa, J. Shu, S. Kaliaguine, P. A. Grandjean, Thin palladium film formation on shot peening modified porous stainless steel substrates, *Ind. Eng. Chem. Res* **1996**, *35*, 973–977.

226 J. Tong, Y. Matsumura, Thin Pd membrane prepared on macroporous stainless steel tube filter by an in-situ multidimensional plating mechanism, *Chem. Commun.* **2004**, *21*, 2460.

227 P. P. Mardilovich, Y. She, Y. H. Ma, M.-H. Rei, Defect-free palladium membranes on porous stainless-steel support, *AIChE J.* **1998**, *44*(2), 310–322.

228 J. Shu, B. P. A. Grandjean, S. Kaliaguine, Gas permeation and isobutane dehydrogenation over very thin Pd/ceramic membranes, *Can. J. Chem. Eng.* **1997**, *75*(4), 712–720.

229 K. L. Yeung, S. C. Christiansen, A. Varma, Palladium composite membranes by electroless plating technique: Relationships between plating kinetics, film microstructure and membrane performance, *J. Membr. Sci.* **1999**, *159*(1-2), 107–122.

230 A. Adrover, M. Giona, L. Capobianco, P. Tripodi, V. Violante, Stress-induced diffusion of hydrogen in metallic membranes: Cylindrical vs. Planar formulation. II, *J. Alloys Compd.* **2003**, *358*, 157–167.

231 W. S. Zhang, M. Q. Hou, H. Y. Wang, Y. B. Fu, Numerical simulation of diffusivity of hydrogen in thin tubular metallic membranes affected by self-stresses, *Int. J. Hydrogen Energy* **2004**, *29*, 1165–1172.

232 P. Zoltowski, Concentration transfer function of hydrogen diffusion in self-stressed metals, *J. Electroanal. Chem.* **2001**, *512*, 64–73.

233 K. Kandasamy, F. A. Lewis, Important Gor*sky* effect influences on diffusion coefficients in metal-hydrogen systems, *Int. J. Hydrogen Energy* **1999**, *24*, 763–769.

234 F. A. Lewis, K. Kandasamy, X. Q. Tong, Lattice strain gradient effects on hydrogen diffusion parameter calculations, *Int. J. Hydrogen Energy* **2002**, *27*, 687–694.

235 X. Q. Tong, Y. Sakamato, F. A. Lewis, R. V. Bucur, K. Kandasamy, "Uphill" hydrogen diffusion effects and hydrogen diffusion coefficients in palladium, *Int. J. Hydrogen Energy* **1997**, *22*(2/3), 141–144.

236 K. Aasberg-Petersen, C. S. Nielsen, S.L. Jørgensen, Membrane reforming for hydrogen, *Catal. Today* **1998**, *46*(2–3), 193–201.

237 B. Emonts, J. Bøgild Hansen, S. Lægsgaard Jørgensen, B. Höhlein, R. Peters, Compact methanol reformer test for fuel-cell powered light-duty vehicles, *J. Power Sources* **1998**, *71*, 288–293.

238 A. Criscuoli, A. Basile, E. Drioli, O. Loiacono, An economic feasibility study for water gas shift membrane reactor, *J. Membr. Sci.* **2001**, *181*, 21–27.